

LOBE-FINNED FISH

Once considered to be the ancestors of land vertebrates, lobe-finned fish have fins that resemble primitive limbs, with a fleshy base preceding the fin membrane.

Like the ray-finned fish, these are bony fish with a hard skeleton but their fins have a different structure. The fin membrane is supported on a muscular lobe that projects out from the body and is strong enough to let some of these fish shuffle along on their paired pectoral and pelvic fins. Bones and cartilage inside the lobes provide attachment for muscles. There are many fossil groups of lobe-finned fish but living representatives are confined to the marine coelacanths and the freshwater lungfish.

A LIVING FOSSIL

Coelacanths are nocturnal and secretive. The first living specimen was discovered in 1938 – until then only fossil species over 65 million years old were known. The historic find belonged to a species living in deep rocky areas in

the western Indian Ocean. A second species was found in Indonesian waters in 1998. They have several unique structural features including an incomplete backbone, residual notochord, and a tail fin with a characteristic extra middle lobe. Their scales are heavy bony plates and they are not long-distance swimmers. Unlike the egg-laying lungfish, coelacanths bear live young as the eggs hatch internally. Gestation may be as long as three years, and this makes their survival precarious if individuals continue to be caught in the nets of deep-sea trawlers.

BREATHING OUT OF WATER

Although most of their fossil ancestors lived in the ocean, modern lungfish are limited to freshwater habitats in South America, Africa, and Australia. All can to some extent breathe air via a connection to the swim bladder – useful when pools dry out seasonally. Some species can survive buried in mud for many months and would die if kept permanently submerged in water, while others still rely mainly on gills for breathing. Their shape and the fact that the larvae of some species have external gills led early zoologists to believe that lungfish were amphibians.

PHYLUM	CHORDATA
CLASSES	COELACANTHI DIPNEUSTI
ORDERS	3
FAMILIES	4
SPECIES	8

DEBATE

FISH ON LAND

While it is generally accepted that land-living vertebrates evolved from a primitive fish or fish-like ancestor in the sea, finding that ancestor is more difficult. Recent work suggests that lungfish are more closely related to tetrapods (four-limbed vertebrates such as mammals) than are coelacanths. In 2002 a fossil lobe-finned fish, *Styloichthys*, was found in China. It appears to show close links between lungfish and tetrapods. The debate continues, but currently coelacanths are not considered direct ancestors of tetrapods.

AFRICAN LUNGFISH

All four species of African lungfish have a long body and thread-like pectoral and pelvic fins, collectively called paired fins. The Lepidosireniformes breathe through a pair of lungs derived from their swim bladder.



2 m
6½ ft

WEST AFRICAN LUNGFISH
Protopterus annectens
F: Protopteridae

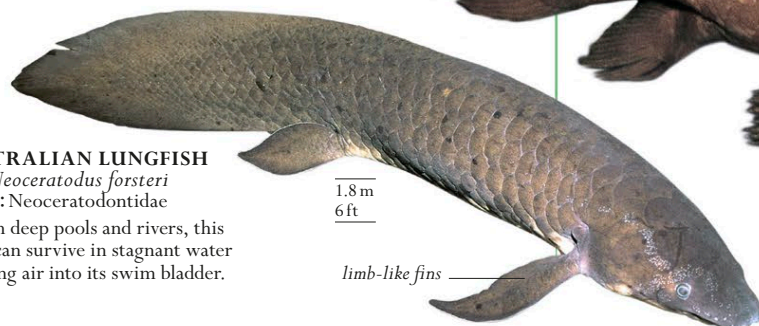
When the lakes in which it lives dry out, this lungfish survives by burying itself in mud, forming a cocoon with an inlet for air.

AUSTRALIAN LUNGFISH

The single species in the order Ceratodontiformes has a long body, large scales, paddle-like paired fins, and a tapering tail. It can breathe through its lungs for short periods, but cannot survive if its habitat dries out.

AUSTRALIAN LUNGFISH
Neoceratodus forsteri
F: Neoceratodontidae

Living in deep pools and rivers, this lungfish can survive in stagnant water by gulping air into its swim bladder.



1.8 m
6 ft

limb-like fins

COELACANTHS

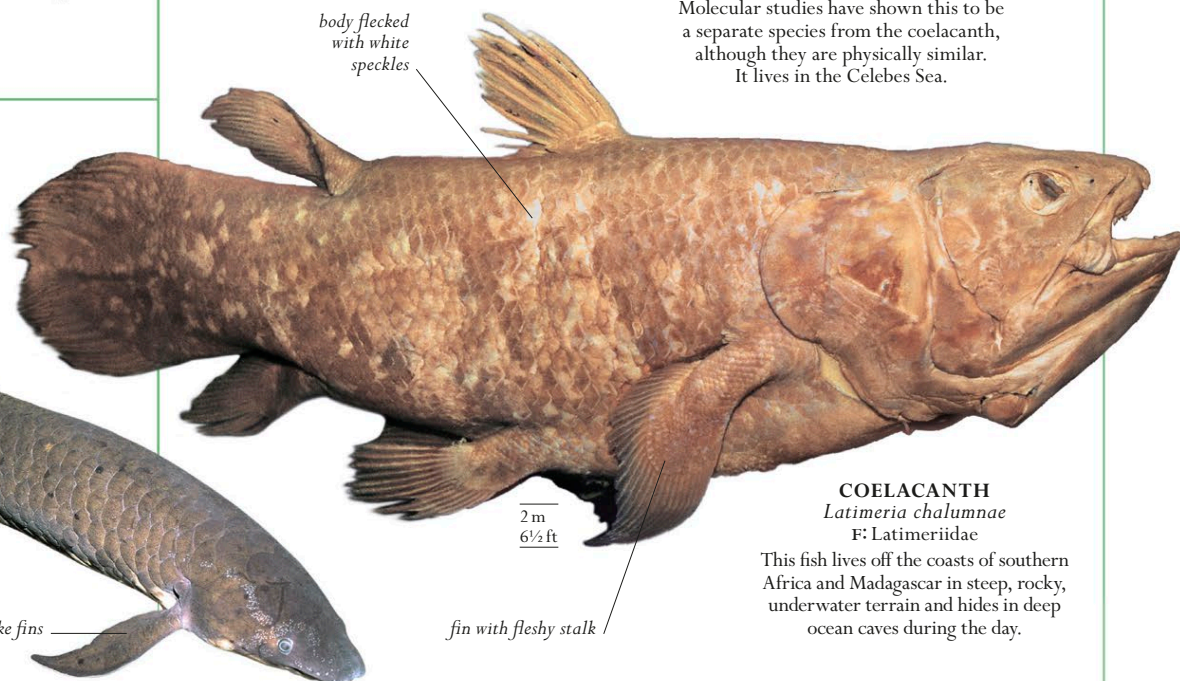
The two primitive fish from the order Coelacanthiformes have limb-like fleshy bases to their pectoral and pelvic fins, and large bony scales. In life, these fish are a metallic blue with pale spots, but the colour fades on death. Interesting 'head-standing' behaviour has been observed from a submersible.



1.4 m
4½ ft

INDONESIAN COELACANTH
Latimeria menadoensis
F: Latimeriidae

Molecular studies have shown this to be a separate species from the coelacanth, although they are physically similar. It lives in the Celebes Sea.



2 m
6½ ft

fin with fleshy stalk

COELACANTH
Latimeria chalumnae
F: Latimeriidae

This fish lives off the coasts of southern Africa and Madagascar in steep, rocky, underwater terrain and hides in deep ocean caves during the day.